

Code-Layout, Readability and Reusability

Date	01 July 2026
Team ID	SWTID-2026-8998
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Improves readability
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static files, models, and datasets
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Easy to understand and maintain
4	Function / Method Names	Functions are descriptively named	Yes	Clearly represents each functionality
5	Code Comments	Inline and block comments explain logic	Yes	Helps future development
6	Modular Design	Code is split into reusable functions	Yes	Frontend, backend, and ML components are separated
7	No Redundant Code	Duplicate or unused code is removed	Yes	Clean implementation
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Uses Flask exception handling and input validation

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application (app.py)	Python, Flask	Used throughout the web application	High
2	Data Preprocessing Module	Python, Pandas	Shared across multiple web pages	High
3	Machine Learning Model	Scikit-learn, XGBoost	Used for real-time credit card a	High
4	HTML Templates	HTML, Bootstrap	Used during training and prediction	Medium
5	CSS Stylesheet	CSS	Handles routing and prediction requests	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate modules
Readability	5	Clear naming conventions and understandable code
Reusability	5	Reusable preprocessing functions, ML model, and Flask routes
Documentation / Comments	5	Sufficient comments and project documentation
Overall Score	5	Code is clean, maintainable, and follows good programming practices